AI-Driven Cloud Platform for Personalized Drug Response Prediction

Multi-omics data · Graph neural network + Transformer · Deployed on AWS

Abhijeet Sinha
23BIT0157

Sarthak Pathak
23BIT0187

Shriyans Pratihari
23BIT0219

BITE412L Cloud Computing · Guide: Dr. Priya V · SCOPE, VIT Vellore · 16/10/2026

Good drug-response models rarely become usable services

Research code, not services

No login, storage, audit trail or monitoring, so a team cannot use them safely

Partial use of the data

One or two omics layers; drugs as strings; missing data not handled

Limited explanations

Deep models are hard to trust in a clinical setting

Optimistic evaluation

Random splits leak cell lines; accuracy drops on unseen data

61

deep-learning drug response models catalogued in one review (Partin et al., 2023)

19

studies found that actually ran an MLOps pipeline in health care (Li et al., 2025)

15 papers (2023–2026) point to one combined gap

Models (8)

P2, P6–P10, P12, P13

- Graph drug encoders (GPDRP, TransCDR)
- Attention / Transformer fusion (PASO, PharmaFormer)
- Cell line → patient transfer (PANCDR)

Reviews (4)

P1, P3, P14, P15

- Big accuracy drops on unseen datasets
- Simple models often match complex ones
- Need validation on new patients and drugs

Cloud & MLOps (3)

P4, P5, P11

- Serverless cuts RNA-seq runtime by up to 96%
- Security and performance are open issues
- Deployment and monitoring rarely done

Gap: no reviewed work delivers an explainable multi-omics model as a secure, monitored, low-cost cloud service evaluated for both accuracy and service quality.

Six measurable objectives

01

Leakage-free preprocessing

Statistics fitted on training cell lines only

02

Multimodal GNN + Transformer

Pearson ≥ 0.80, ROC-AUC ≥ 0.85 on unseen cell lines

03

Explainable predictions

Modality + gene attributions with every result

04

Secure deployment as code

One CloudFormation stack; access tests pass

05

Performance & observability

Median latency ≤ 2 s; metrics and alarms live

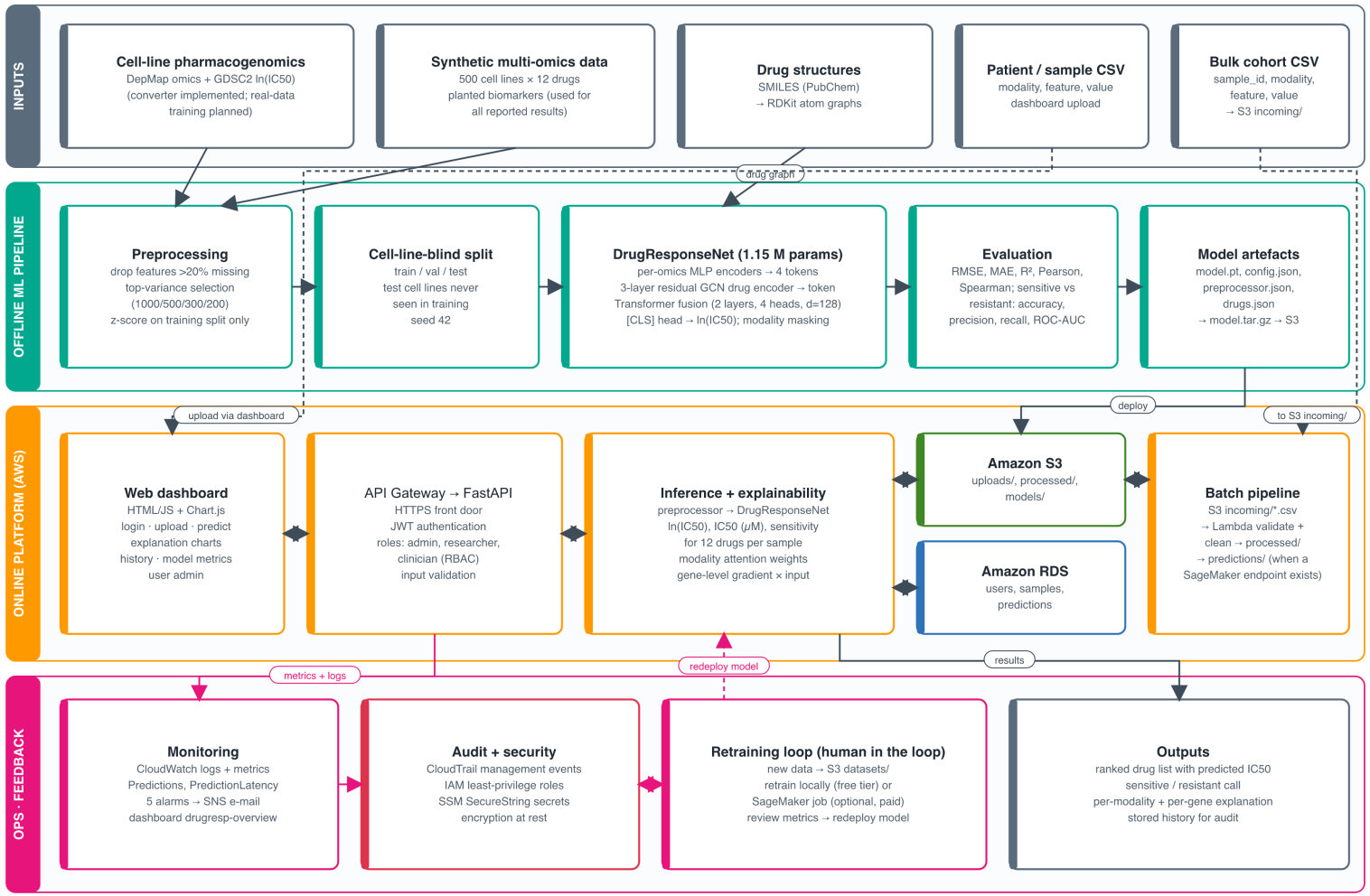
06

Cost control

AWS Free plan; ≤ US$50/month on-demand equivalent

From multi-omics profile to explained prediction

Figure 2. Complete System Architecture and Workflow



1. Upload a profile in the dashboard
2. API authenticates and stores it in S3
3. Model predicts ln IC50 for 12 drugs
4. Explanations + results saved to RDS
5. CloudWatch tracks every prediction

DrugResponseNet: one token per data type

Expression

1,000 genes

Copy number

500 genes

Mutation

300 genes

Proteomics

200 proteins

Drug graph

RDKit atoms → 3-layer GCN

Transformer fusion

- [CLS] token attends to all tokens
- 2 layers · 4 heads · d = 128
- Missing modalities masked out
- Modality dropout 0.15 in training

1.15 M parameters

Output

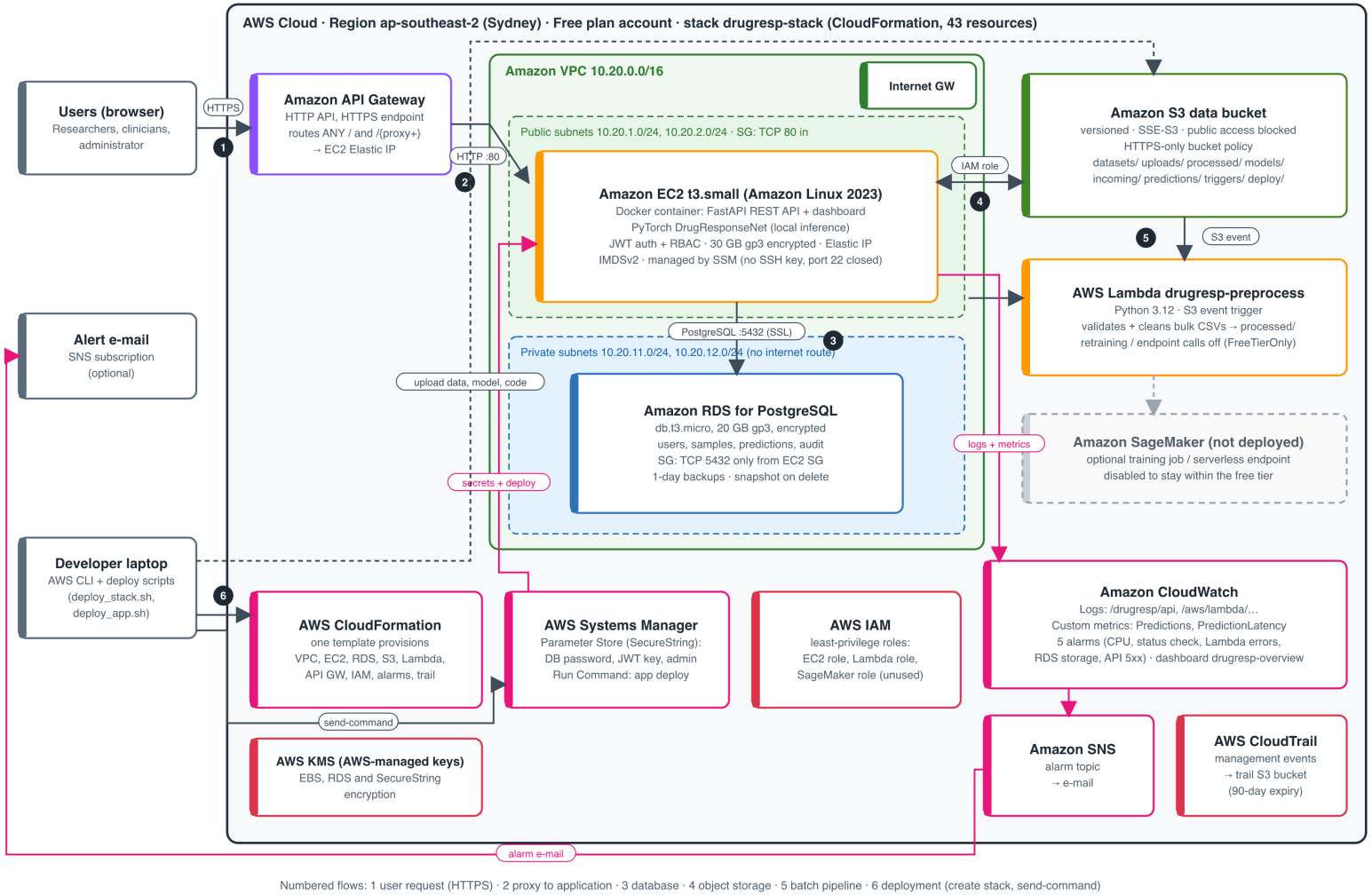
ln IC50 and IC50 (µM) per drug; sensitive vs resistant against the drug's training median

Explanations

Attention weights show which data type mattered; gradient × input ranks the top genes

Deployed on AWS from one CloudFormation template

Figure 1. Cloud Deployment Architecture (as deployed, AWS ap-southeast-2)



Numbered flows: 1 user request (HTTPS) · 2 proxy to application · 3 database · 4 object storage · 5 batch pipeline · 6 deployment (create stack, send-command)

43

resources in stack drugresp-stack

11

AWS services in use

3

commands to deploy

Security is built into the stack



Private database

RDS in private subnets; port 5432 only from the app



Login and roles

HTTPS, JWT, admin / researcher / clinician, sample ownership



No secrets in code

SecureString parameters in SSM Parameter Store



No SSH

Managed through SSM Run Command; IMDSv2 enforced



Encrypted everywhere

S3 SSE + HTTPS-only policy; encrypted EBS and RDS



Audit and alarms

CloudTrail events; 5 CloudWatch alarms → SNS

Data: synthetic benchmark with planted biology

500	12	5,101	2,000
cell lines, 8 lineages	GDSC drugs	response pairs	selected omics features

Cell-line-blind split: 3,566 / 763 / 772 pairs; test cell lines never seen in training

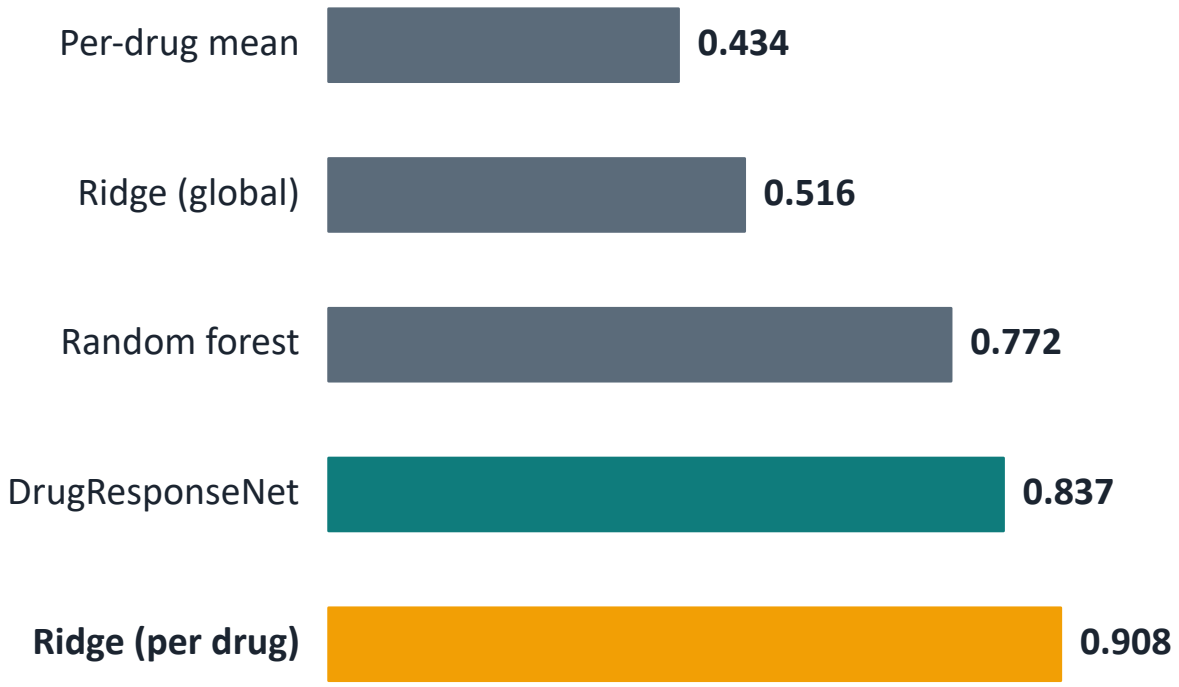
Leakage-free: feature selection and z-scores fitted on training cell lines only

Planted effects: EGFR → erlotinib, BRCA1 → olaparib, ABCB1 → doxorubicin resistance

Real data next: DepMap + GDSC converter and TCGA validation are implemented but not yet run

Beats global baselines; a per-drug linear model scores higher

Test Pearson r on 772 pairs from unseen cell lines



Same split, features and thresholds for every model

0.837	0.866	1.077
Pearson r	ROC-AUC	RMSE (ln IC50)

Why the per-drug ridge wins

The synthetic data plant mostly linear, drug-specific effects, which a separate linear model per drug captures exactly. Our single model must also handle all drugs, missing data and new drugs from structure. Real DepMap/GDSC data and an unseen-drug test will decide.

Live deployment: 13 of 14 tests passed

Area	Tests	What was checked	Pass
Functional	T01–T04	12 unit tests; stack created; health OK; end-to-end prediction	**4/4**
Security	T05–T08, T11	No token, forged token, wrong password → 401; sign-up → 403; SSH closed	**5/5**
Access control	T09–T10	Researcher blocked from user admin (403) and others' samples (404)	**2/2**
Performance	T13–T14	Latency and 10-client concurrency within targets	**2/2**
HTTPS bypass	T12	EC2 answers plain HTTP on its public IP; fix: restrict port 80	**0/1**

1.07 s

median end-to-end, 12 drugs + explanations

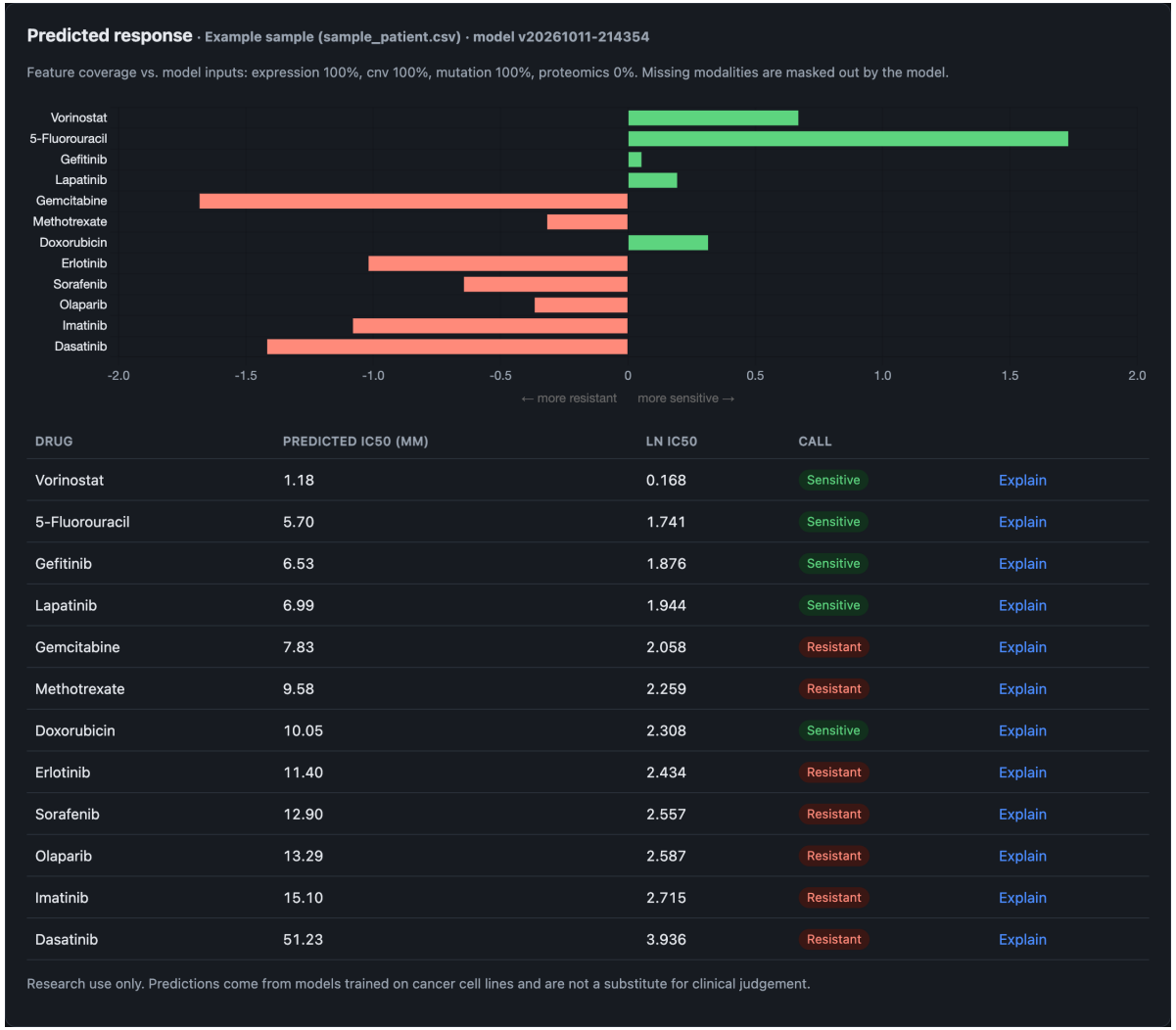
0.18 s

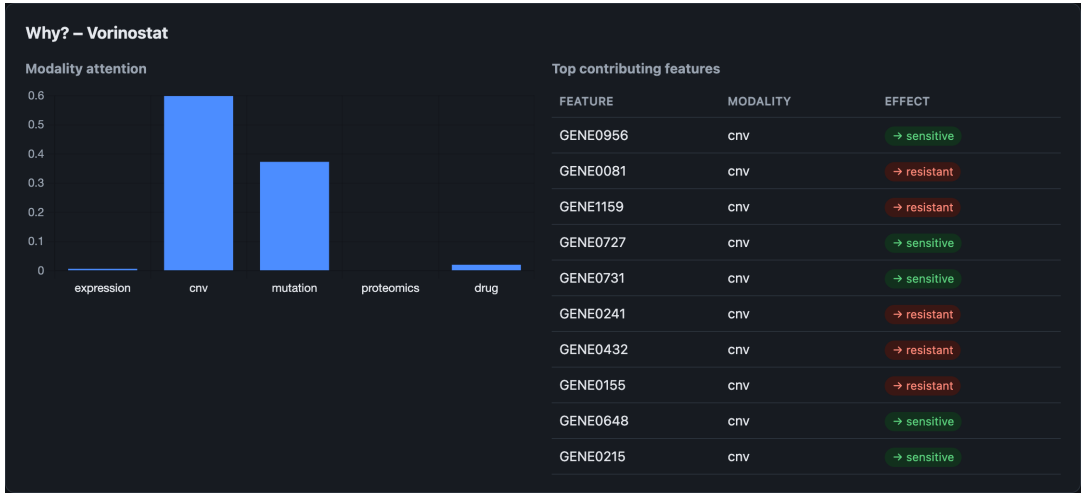
average server-side inference (CloudWatch)

6.6 / s

predictions with 10 concurrent users, 0 errors

The dashboard: ranked drugs with explanations

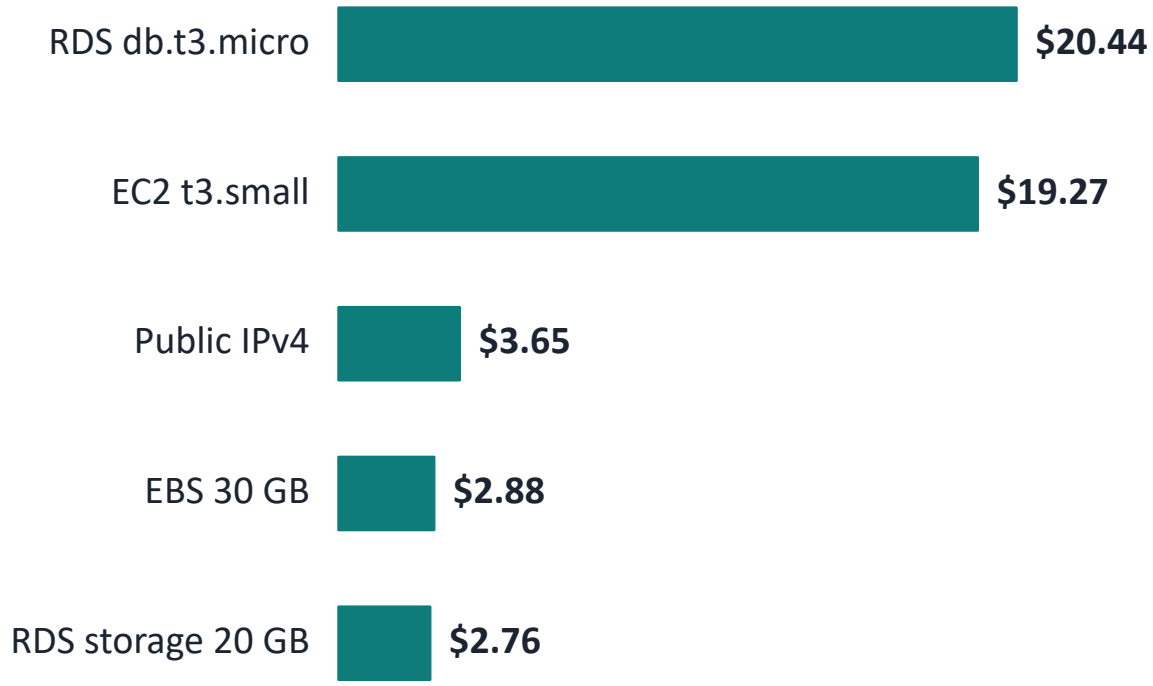




Left: predicted sensitivity for 12 drugs with IC50 and sensitive/resistant calls. Right: why vorinostat was predicted sensitive, by data type and top genes.

Runs on the AWS Free plan: no card charges

Estimated on-demand cost, ap-southeast-2 (USD per month)



≈ $49

per month if billed on demand (AWS price lists, 16/10/2026); everything else < $0.10

Cost controls

- Free plan: usage comes from credits
- SageMaker switched off (FreeTierOnly)
- No NAT gateway; Parameter Store, not Secrets Manager
- Stop EC2/RDS between sessions; delete stack after

Limitations and next steps

Current limitations

- Synthetic data only; one training run
- Per-drug ridge beats the model here
- Port 80 reachable without HTTPS (T12)
- Single EC2 instance, Single-AZ RDS
- Region fixed to Sydney (extra latency)

Next steps

- Train on DepMap + GDSC; test unseen drugs; run TCGA validation
- Repeat runs with several seeds; report confidence intervals
- Restrict port 80 to API Gateway or use a private VPC link
- Auto Scaling across two AZs; Multi-AZ RDS
- Move to Mumbai; add CI/CD and SNS e-mail alerts

Team contributions



Abhijeet Sinha

23BIT0157

- Data, preprocessing and DrugResponseNet
- Training, metrics and baselines
- Papers P2, P7, P10, P12, P13



Sarthak Pathak

23BIT0187

- CloudFormation, IAM, VPC, SSM
- FastAPI backend, RBAC, PostgreSQL
- Papers P1, P4, P5, P11, P14



Shriyans Pratihari

23BIT0219

- Dashboard and API integration
- End-to-end and UI testing
- Papers P3, P6, P8, P9, P15

An explainable drug-response model, delivered as a secure cloud service for $0

O1, O3, O5, O6 met · O4 met with one finding · O2 partly met

Code: github.com/Vegeta290/AIDrugResponsePrediction_AWS_Cloud_Project_2026

Live: q6oovdd9rc.execute-api.ap-southeast-2.amazonaws.com

Thank you · Questions?